

Qualification Pack



Energy Meter Technician

QP Code: PSS/Q2504

Version: 1.0

NSQF Level: 3

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PSS/Q2504: Energy Meter Technician

Brief Job Description

An energy meter technician, who is also known as consumer energy meter technician, performs installation and commissioning of conventional meters as well as smart meters in line with standard work practices.

Personal Attributes

The candidate should be Physically and mentally able to safely perform essential functions of the job, climb ladders, scaffolds and poles of various heights. The candidate should be able to read, hear and understand instructions and warnings.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N2509: Manually remove, change and install Low voltage, single and three phase energy meters](#)
2. [PSS/N2508: Install and commission smart meter](#)
3. [PSS/N1331: Apply basic health and safety practices for power related work](#)
4. [PSS/N1336: Working effectively with others](#)
5. [SGJ/N1702: Optimize resource utilization at workplace](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution
Country	India
NSQF Level	3
Credits	NA



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Aligned to NCO/ISCO/ISIC Code	NCO-2015/3113.9900
Minimum Educational Qualification & Experience	10th Class with NA of experience
Minimum Level of Education for Training in School	9th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	14/04/2020
Next Review Date	14/04/2023
NSQC Approval Date	22/09/2020
Version	1.0
Reference code on NQR	2020/POW/PSSC/03905
NQR Version	1.0



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PSS/N2509: Manually remove, change and install Low voltage, single and three phase energy meters

Description

This NOS is about installing, removing or changing, testing and maintaining Low Voltage (LV) consumer energy meters (single phase or three phase) and meter supportive equipment that are used to record energy consumption in residential, commercial or industrial units.

Scope

The scope covers the following :

- Work safely
- Prepare work area for installation
- Install a single or three phase meters
- Remove and replace a single or three phase meter
- Perform meter recording procedures post installation

Elements and Performance Criteria

Work safely

To be competent, the user/individual on the job must be able to:

- PC1.** obtain job specification or work order from responsible authority
- PC2.** select and use appropriate personal protective equipment (PPE) as per occupational health and safety guidelines (Personal protective equipment: hard working caps, protective glasses, rubber gloves, fall arrest and restraint, safety footwear, fire-resistant clothing, etc.)
- PC3.** use and maintain appropriate tools and equipment in accordance with the tasks (Tools and equipment: e.g. insulated hand tools; drills; hacksaw; hand tools; testing equipment; insulation testers; crimping tools; wires and cables of various colours and sizes; heat shrink sleeving and flexible conduit; terminals and connectors; electrical tape; etc.)

Prepare work area for installation

To be competent, the user/individual on the job must be able to:

- PC4.** verify the distance between the poles or cables is as per standards
- PC5.** check the underground and/or overhead cables are laid correctly as per work order
- PC6.** plan and locate the area inside or outside the customer's premise after assessing possible risks
- PC7.** check that the identified area is accessible to carry out installation, meter testing, commissioning, reading, recording and maintenance
- PC8.** ensure the energy meter has passed quality assessments set by the Bureau of Indian Standards (BIS) and is ready for installation (Consumer meters: Low Voltage(LV) meters; single phase meter (two wires system) and three phase meters (four wires system))
- PC9.** inspect the facility's wiring system and recognize any possible risks to be isolated or common phase or looping of phase of two or more consumers



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PC10. inform all affected parties of the intended work plan in advance prior to disconnecting power supply line

Install single or three phase meters

To be competent, the user/individual on the job must be able to:

PC11. install the energy meter and supportive equipment using insulated tools and devices as per organizational procedures (Supportive equipment: e.g. meter box, junction box, distribution bus bar, etc.)

PC12. equip the energy meter with various anti-tampering features as per regulations and organizational procedures

PC13. establish immunity against various types of external factors in accordance with relevant regulations (External factors: magnetic induction, vibration, electrostatic discharge, switching transients, surge voltages, oblique suspension and harmonics)

PC14. ensure the energy meter displays one or more of the following parameters depending upon the tariff requirement for different categories of consumers (Measuring parameters: cumulative reactive energy, average power factor, time of energy use, apparent power, maximum demand, phase voltage and line currents)

PC15.

- check the energy meter for earth leakage indication as per relevant regulations
- Regulations: Central Electricity Authority Regulations, 2006

PC16. test and calibrate the energy meter using appropriate testing devices in line with organizational quality standards and regulations

PC17. identify and escalate unresolved problems to appropriate authority for rectifications

Remove and replacing a single or a three phase meter

To be competent, the user/individual on the job must be able to:

PC18.

- establish the reason for changing the energy meter from responsible source in order to plan out the work
- Reasons: discrepancies (stoppage of meter, erratic consumption output, broken seal, burning or damage of meter), service disconnection

PC19. replace the same with a duly tested energy meter as per instructions given in organizational guidelines and regulations

PC20. test to confirm that the replaced energy meter conforms to required work specifications

PC21. check that any replaced or repaired equipment is working properly, and customer's problems are resolved efficiently

Meter recording procedures post installation

To be competent, the user/individual on the job must be able to:

PC22. record and verify the metered data and maintain all the information related to the consumer's energy meter

PC23. maintain consumer meters' account history, installation date and testing details, calibration and replacement of meters in line with organizational standards and policies

PC24. check that tools and devices used are disassembled and stored safely as per instructions

PC25. dispose waste materials such as wires, tapes, plastic caps, etc. in line with safety and environmental procedure

PC26. clear the work area of hazards and hazardous substances

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** relevant standards, legislation, working procedures and policies of organization relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the workplace
- KU3.** importance of working in clean and safe environment
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** documentation and related procedures applicable in the context of employment and work
- KU6.** installation, operation and maintenance procedures of energy meter as listed under the Central Electricity Regulations, 2006
- KU7.** job specification documents, their sources and interpretation
- KU8.** work planning process.
- KU9.** safety control measures such as signs and barriers, demarcation of work area, control and removal of hazards and contamination protection
- KU10.** list of required tools and equipment and their uses in the work
- KU11.** various types of consumer energy meters, their components and their uses (Types of meters: single phase meter, three phase meter, CT meter and HT meters)
- KU12.** quality check parameters for consumer energy meters such as specification of meters, immunity to external factors, sealing points and functional requirements, etc.
- KU13.** meter specifications such as standard reference voltage, voltage range, standard frequency, standard basic current, accuracy class, starting current and maximum current, power factor range, power frequency withstand voltage, impulse voltage withstand test for 1.2/50 micro sec, power consumption check tampering, breaking or removing
- KU14.** compliance with energy meter standards set by apex regulators (Regulators: Bureau of Indian Standards (BIS), British Standards (BS), International Electro-technical Commission (IEC) Standards, etc.)
- KU15.** how to select suitable location for installing an energy meter
- KU16.** application of basic principles of electricity in energy meters
- KU17.** electrical units used to measure energy outputs, for example KVA, KWH, etc.
- KU18.** metered data and information database to be maintained and data verification procedure
- KU19.** energy meters testing procedures and devices used
- KU20.** standard features of a correct energy meter as defined by regulating body and required meter specifications as per Indian Standards
- KU21.** various anti-tampering features in an energy meter
- KU22.** correct waste disposal methods against safety and environmental issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record technical information such as observations and key findings
- GS2.** fill logs, forms and formats in local or English language for administrative purpose, quality related information, service related information, etc. whenever needed



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- GS3.** interpret pictorial or graphical representations and written signs or instructions on electrical units
- GS4.** read and correctly interpret rules and methods stated in work related documents such as job specifications and equipment manuals
- GS5.** discuss task lists, schedules and activities with the concerned authority
- GS6.** communicate clearly with the team/group members
- GS7.** make appropriate work-related judgments based on policy and procedures
- GS8.** follow organization's rule-based decision-making process
- GS9.** plan and organise tasks to meet deadlines
- GS10.** plan maintenance work to ensure it is completed within specified time
- GS11.** build customer relationships and use customer centric approach in taking decisions

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work safely</i>	3	7	-	-
PC1. obtain job specification or work order from responsible authority	1	2	-	-
PC2. select and use appropriate personal protective equipment (PPE) as per occupational health and safety guidelines (Personal protective equipment: hard working caps, protective glasses, rubber gloves, fall arrest and restraint, safety footwear, fire-resistant clothing, etc.)	1	2	-	-
PC3. use and maintain appropriate tools and equipment in accordance with the tasks (Tools and equipment: e.g. insulated hand tools; drills; hacksaw; hand tools; testing equipment; insulation testers; crimping tools; wires and cables of various colours and sizes; heat shrink sleeving and flexible conduit; terminals and connectors; electrical tape; etc.)	1	3	-	-
<i>Prepare work area for installation</i>	7	18	-	-
PC4. verify the distance between the poles or cables is as per standards	1	2	-	-
PC5. check the underground and/or overhead cables are laid correctly as per work order	1	2	-	-
PC6. plan and locate the area inside or outside the customer's premise after assessing possible risks	1	2	-	-
PC7. check that the identified area is accessible to carry out installation, meter testing, commissioning, reading, recording and maintenance	1	3	-	-
PC8. ensure the energy meter has passed quality assessments set by the Bureau of Indian Standards (BIS) and is ready for installation (Consumer meters: Low Voltage(LV) meters; single phase meter (two wires system) and three phase meters (four wires system))	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. inspect the facility's wiring system and recognize any possible risks to be isolated or common phase or looping of phase of two or more consumers	1	3	-	-
PC10. inform all affected parties of the intended work plan in advance prior to disconnecting power supply line	1	3	-	-
<i>Install single or three phase meters</i>	10	25	-	-
PC11. install the energy meter and supportive equipment using insulated tools and devices as per organizational procedures (Supportive equipment: e.g. meter box, junction box, distribution bus bar, etc.)	2	6	-	-
PC12. equip the energy meter with various anti-tampering features as per regulations and organizational procedures	2	3	-	-
PC13. establish immunity against various types of external factors in accordance with relevant regulations (External factors: magnetic induction, vibration, electrostatic discharge, switching transients, surge voltages, oblique suspension and harmonics)	1	3	-	-
PC14. ensure the energy meter displays one or more of the following parameters depending upon the tariff requirement for different categories of consumers (Measuring parameters: cumulative reactive energy, average power factor, time of energy use, apparent power, maximum demand, phase voltage and line currents)	1	3	-	-
PC15. - check the energy meter for earth leakage indication as per relevant regulations - Regulations: Central Electricity Authority Regulations, 2006	1	3	-	-
PC16. test and calibrate the energy meter using appropriate testing devices in line with organizational quality standards and regulations	2	4	-	-
PC17. identify and escalate unresolved problems to appropriate authority for rectifications	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Remove and replacing a single or a three phase meter</i>	5	9	-	-
PC18. - establish the reason for changing the energy meter from responsible source in order to plan out the work - Reasons: discrepancies (stoppage of meter, erratic consumption output, broken seal, burning or damage of meter), service disconnection	1	2	-	-
PC19. replace the same with a duly tested energy meter as per instructions given in organizational guidelines and regulations	1	3	-	-
PC20. test to confirm that the replaced energy meter conforms to required work specifications	2	2	-	-
PC21. check that any replaced or repaired equipment is working properly, and customer's problems are resolved efficiently	1	2	-	-
<i>Meter recording procedures post installation</i>	5	11	-	-
PC22. record and verify the metered data and maintain all the information related to the consumer's energy meter	1	3	-	-
PC23. maintain consumer meters' account history, installation date and testing details, calibration and replacement of meters in line with organizational standards and policies	1	2	-	-
PC24. check that tools and devices used are disassembled and stored safely as per instructions	1	2	-	-
PC25. dispose waste materials such as wires, tapes, plastic caps, etc. in line with safety and environmental proced	1	2	-	-
PC26. clear the work area of hazards and hazardous substances	1	2	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2509
NOS Name	Manually remove, change and install Low voltage, single and three phase energy meters
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	NA
Next Review Date	NA
NSQC Clearance Date	NA



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PSS/N2508: Install and commission smart meter

Description

This unit is about installing and commissioning smart meters as per standard work practices.

Scope

The scope covers the following :

- Prepare for installation of smart meter
- Install smart meter
- Commission smart meter

Elements and Performance Criteria

Prepare for installation of smart meter

To be competent, the user/individual on the job must be able to:

- PC1.** identify the Advanced Metering Infrastructure (AMI) to be installed; such as smart meters (single-phase and 3-phase), communication infrastructure, Head-End System (HES), Meter Data Management System (MDMS)
- PC2.** determine the communication network/technology to be used such as Broadband over PowerLine (BPL), Power Line Communications (PLCs), Fiber Optic Communication, Zigbee, 6LoWPAN, Bluetooth, Fixed Radio Frequency or public networks (e.g., landline, cellular, paging)
- PC3.** evaluate key factors for selecting an appropriate installation site such as network reliability, availability of existing network; convenient location for the user/customer; required signal strength etc.
- PC4.** confirm the availability of required tools, equipment, materials, hardware and software for installing smart meter

Install smart meter

To be competent, the user/individual on the job must be able to:

- PC5.** install smart meter box and smart metering hardware
- PC6.** perform wiring and cable connections
- PC7.** install the In-Home Display (IHD)
- PC8.** configure smart meter to establish required connectivity with the Head-End System (HES)
- PC9.** enter the IP address as provided through the HES for enabling networking capabilities of the smart meter

Commission smart meter

To be competent, the user/individual on the job must be able to:

- PC10.** test the smart meter for potential hardware issues; such as orientation of meter; improper connections; using a different port instead of recommended etc.
- PC11.** check the connections, features, switches within the smart metering network for required functioning
- PC12.** test the In-Home Display (IHD) for required connectivity and functioning



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- PC13.** check the communication network (Wireless Area Network (WAN), Home Area Network (HAN) etc.) to evaluate two-way communication capability between smart meter and HES
- PC14.** verify that the smart meter cover is securely fitted to ensure adequate protection against extreme environmental conditions
- PC15.** record data such as customer identification number, meter ID, its hardware and software configuration, name plate details, make, type i.e. 1 Phase or 3 Phase, etc. (as per requirement of utility)
- PC16.** prepare work reports for the tasks performed
- PC17.** instruct the users/customers the various procedures through which they can avail of customer services offered. Customer services: viewing meter readings and reports; updating personal information such as mobile number, email; generating service requests; viewing recharge history, present balance etc.
- PC18.** clear the worksite to ensure no hazardous and waste materials are present

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the organisation
- KU2.** relevant health and safety requirements applicable in the workplace
- KU3.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU4.** basic electrical concepts such as power factor, current and voltage transmission
- KU5.** various symbols and terminologies used in smart metering systems
- KU6.** role of various elements of smart metering system such as HES, communication platform, MDMS etc.
- KU7.** networking for smart meter installation such as configuring IP address, implementation of networking topologies such as HAN (Home Area Network), WAN (Wide Area Network) and MAN (Metropolitan Area Network), establishing communication, etc.
- KU8.** basics of communication technologies such as RF, GSM/CDMA, fiber optic communications, PLCs, Zigbee, 6LoWPAN, Bluetooth etc.
- KU9.** identification of different ports used in the installation of smart meters such as RS 232, RS 485 etc.
- KU10.** importance of job specifications in the installation and commissioning process
- KU11.** considerations for selecting an appropriate location for smart meter installation
- KU12.** tools, equipment, hardware/software technologies, wiring, cabling and other materials required for smart meter installation and commissioning
- KU13.** installation of the In-Home Display (IHD) in an appropriate location
- KU14.** procedures for various checks and tests. Checks and tests: Testing the functionalities of smart meter, checking basic connections, features, switches within the smart metering network; testing of the IHD as per required standards; checking of the communication capabilities between HES and smart meter
- KU15.** procedure to provide adequate protection to the smart meter
- KU16.** problematic issues and situations that require expert intervention
- KU17.** document necessary information such as observations, work reports, key findings, etc.



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KU18. relevant standards/regulations/prevaling practices of the utility for smart meter installation and commissioning

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record technical information such as observations and key findings
- GS2.** fill logs, forms and formats in local or English language for administrative purpose, quality related information, service related information, etc. whenever needed
- GS3.** interpret pictorial or graphical representations and written signs or instructions on electrical units
- GS4.** read and correctly interpret rules and methods stated in work related documents such as job specifications and equipment manuals
- GS5.** discuss task lists, schedules and activities with the concerned authority
- GS6.** communicate clearly with the team/group members
- GS7.** make appropriate work-related judgments based on policy and procedures
- GS8.** follow organization's rule-based decision-making process
- GS9.** plan and organise tasks to meet deadlines
- GS10.** plan maintenance work to ensure it is completed within specified time
- GS11.** build customer relationships and use customer centric approach in taking decisions

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for installation of smart meter</i>	5	10	-	-
PC1. identify the Advanced Metering Infrastructure (AMI) to be installed; such as smart meters (single-phase and 3-phase), communication infrastructure, Head-End System (HES), Meter Data Management System (MDMS)	1	2	-	-
PC2. determine the communication network/technology to be used such as Broadband over PowerLine (BPL), Power Line Communications (PLCs), Fiber Optic Communication, Zigbee, 6LoWPAN, Bluetooth, Fixed Radio Frequency or public networks (e.g., landline, cellular, paging)	1	3	-	-
PC3. evaluate key factors for selecting an appropriate installation site such as network reliability, availability of existing network; convenient location for the user/customer; required signal strength etc.	2	2	-	-
PC4. confirm the availability of required tools, equipment, materials, hardware and software for installing smart meter	1	3	-	-
<i>Install smart meter</i>	10	28	-	-
PC5. install smart meter box and smart metering hardware	2	6	-	-
PC6. perform wiring and cable connections	2	5	-	-
PC7. install the In-Home Display (IHD)	2	6	-	-
PC8. configure smart meter to establish required connectivity with the Head-End System (HES)	2	6	-	-
PC9. enter the IP address as provided through the HES for enabling networking capabilities of the smart meter	2	5	-	-
<i>Commission smart meter</i>	15	32	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. test the smart meter for potential hardware issues; such as orientation of meter; improper connections; using a different port instead of recommended etc.	2	5	-	-
PC11. check the connections, features, switches within the smart metering network for required functioning	2	5	-	-
PC12. test the In-Home Display (IHD) for required connectivity and functioning	2	5	-	-
PC13. check the communication network (Wireless Area Network (WAN), Home Area Network (HAN) etc.) to evaluate two-way communication capability between smart meter and HES	2	5	-	-
PC14. verify that the smart meter cover is securely fitted to ensure adequate protection against extreme environmental conditions	2	3	-	-
PC15. record data such as customer identification number, meter ID, its hardware and software configuration, name plate details, make, type i.e. 1 Phase or 3 Phase, etc. (as per requirement of utility)	2	3	-	-
PC16. prepare work reports for the tasks performed	1	2	-	-
PC17. instruct the users/customers the various procedures through which they can avail of customer services offered. Customer services: viewing meter readings and reports; updating personal information such as mobile number, email; generating service requests; viewing recharge history, present balance etc.	1	2	-	-
PC18. clear the worksite to ensure no hazardous and waste materials are present	1	2	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2508
NOS Name	Install and commission smart meter
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	NA
Next Review Date	NA
NSQC Clearance Date	NA



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PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)
- PC3.** use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment
- PC4.** follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)
- PC5.** ensure positive isolation of electrical equipment and system as per the given standards
- PC6.** identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.
- PC7.** use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)
- PC8.** use proper scaffolds, elevated platforms and ladders for working at height
- PC9.** inspect various tools and plants (T & P) routinely for any signs of oil, water leakages
- PC10.** store flammable materials and machine lubricating oil safely and correctly



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PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO₂, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & write an accident/ incident report in local language and/or English
- GS2.** read and comprehend basic content to read labels, charts, signages, manuals etc.
- GS3.** question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions
- GS4.** remain congenial while discussing and debating issues with co-workers
- GS5.** observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS6.** plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS7.** build customer relationships and use customer centric approach
- GS8.** seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	25	36	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
<i>Follow fire safety practices</i>	6	12	-	-
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	13	-	-
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	39	61	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1.5
Version	2.0
Next Review Date	31/08/2026



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accept information and instructions from the supervisor and fellow workers and seek clarification when required
- PC2.** pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt
- PC3.** help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency
- PC4.** use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism
- PC5.** listen attentively and respond appropriately while interacting with others at work
- PC6.** act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)
- PC7.** escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC8.** respect gender diversity in the workplace
- PC9.** use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive
- PC10.** communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC11.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** key elements of active listening and assertive communication
- KU6.** importance of teamwork in organizational and individual success
- KU7.** ethics and disciplined behavior for professional success
- KU8.** expressing and addressing grievances appropriately and managing interpersonal conflict effectively
- KU9.** harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU10.** organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS2.** read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
- GS3.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
- GS4.** display active listening skills while interacting with co-workers and other in the workplace
- GS5.** follow organization's rule-based decision-making process
- GS6.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS7.** meet customer requirements
- GS8.** critically evaluate operation parameters in relation to system normality

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	18	32	-	-
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
<i>Respect diversity</i>	16	24	-	-
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	34	56	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	2.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

SGJ/N1702: Optimize resource utilization at workplace

Description

This unit is about adopting sustainable practices and optimizing use of resources, especially material, energy and waste, in day-to-day operations at work

Scope

The scope covers the following :

- Material conservation practices
- Energy/electricity conservation practices
- Effective waste management/recycling practices

Elements and Performance Criteria

Material conservation practices

To be competent, the user/individual on the job must be able to:

- PC1.** identify ways to optimize usage of material including water in various tasks/activities/processes
- PC2.** check for spills/leakages in various tasks/activities/processes
- PC3.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC4.** carry out routine cleaning of tools, machines and equipment

Energy/electricity conservation practices

To be competent, the user/individual on the job must be able to:

- PC5.** identify ways to optimize usage of electricity/energy in various tasks/activities/processes
- PC6.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC7.** report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment
- PC8.** ensure electrical equipment and appliances are properly connected and turned off when not in use

Effective waste management/recycling practices

To be competent, the user/individual on the job must be able to:

- PC9.** identify recyclable and non-recyclable, and hazardous waste generated
- PC10.** segregate waste into different categories
- PC11.** dispose non-recyclable waste appropriately
- PC12.** deposit recyclable and reusable material at identified location
- PC13.** follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** potential hazards, risks and threats based on the nature of work
- KU2.** layout of the workstation and electrical and thermal equipment used
- KU3.** organizations procedures for minimizing waste
- KU4.** efficient and inefficient utilization of material and water
- KU5.** ways of efficiently managing material and water in the process
- KU6.** basics of electricity and prevalent energy efficient devices
- KU7.** ways to recognize common electrical problems
- KU8.** common practices of conserving electricity
- KU9.** usage of different colours of dustbins
- KU10.** categorization of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU11.** waste management and methods of waste disposal
- KU12.** common sources of pollution and ways to minimize it

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data on waste disposal at workplace
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** read Standard Operating Practices (SOP) documents
- GS4.** communicate with colleagues on the significance of greening of jobs
- GS5.** make timely decisions for efficient utilization of resources
- GS6.** complete tasks efficiently and accurately within stipulated time
- GS7.** work with supervisors/team members to carry out work related tasks
- GS8.** identify cause and effect of greening of jobs

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Material conservation practices</i>	4	8	-	-
PC1. identify ways to optimize usage of material including water in various tasks/activities/processes	1	2	-	-
PC2. check for spills/leakages in various tasks/activities/processes	1	2	-	-
PC3. plug spills/leakages and escalate to appropriate authority if unable to rectify	1	2	-	-
PC4. carry out routine cleaning of tools, machines and equipment	1	2	-	-
<i>Energy/electricity conservation practices</i>	4	8	-	-
PC5. identify ways to optimize usage of electricity/energy in various tasks/activities/processes	1	2	-	-
PC6. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	1	2	-	-
PC7. report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment	1	2	-	-
PC8. ensure electrical equipment and appliances are properly connected and turned off when not in use	1	2	-	-
<i>Effective waste management/recycling practices</i>	5	10	-	-
PC9. identify recyclable and non-recyclable, and hazardous waste generated	1	2	-	-
PC10. segregate waste into different categories	1	2	-	-
PC11. dispose non-recyclable waste appropriately	1	2	-	-
PC12. deposit recyclable and reusable material at identified location	1	2	-	-
PC13. follow processes specified for disposal of hazardous waste	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	13	26	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1702
NOS Name	Optimize resource utilization at workplace
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Resource Optimization
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	25/11/2021
Next Review Date	19/06/2025
NSQC Clearance Date	25/11/2021

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 50% of aggregate marks to successfully clear the assessment.



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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N2509.Manually remove, change and install Low voltage, single and three phase energy meters	30	70	-	-	100	30
PSS/N2508.Install and commission smart meter	30	70	-	-	100	30
PSS/N1331.Apply basic health and safety practices for power related work	39	61	0	0	100	15
PSS/N1336.Working effectively with others	34	56	0	0	90	5
SGJ/N1702.Optimize resource utilization at workplace	13	26	0	0	39	20
Total	146	283	0	0	429	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.